

A Counterexample to G -RNP $\Rightarrow$ RNP

Candidate counterexample packet for Problem 8.2

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Abstract

Dantas, Doucha, Jung and Raunig asked whether the group-equivariant Radon–Nikodym property implies the classical Radon–Nikodym property. The supplied finite-group proof is correct. Its induction idea can in fact be made to work for every locally compact σ -compact group. There is also an elementary observation, apparently missed in the source paper, that its printed notion of weak G -dentability is exactly ordinary dentability. The unrestricted implication nevertheless fails. Let $G = \text{Homeo}_+([0, 1])$ and let it act by composition on $X = C_0((0, 1))$. A theorem of Megrelishvili says that every continuous isometric representation of G on a reflexive Banach space is trivial. The Mazur map then shows that every continuous measure-algebra action allowed in the definition of the G -RNP is trivial. Since $X^G = \{0\}$, X has the G -RNP, while X fails the classical RNP because it contains an isometric copy of c_0 . Hence $G\text{-RNP} \Rightarrow \text{RNP}$ is false for general topological groups.

1 Packet status and source evidence

This packet upgrades and absorbs the earlier finite-group partial packet, which is preserved in the local history directory. The supplied TeX/PDF report is stored in a dated evidence directory.

The source target is Problem 8.2 of Dantas–Doucha–Jung–Raunig, asking whether the dotted implications in their summary diagram hold. The relevant source crop is:

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Problem 8.2. Decide whether the implications of the dotted arrows hold true.

One of the most mysterious implications is the one between the classical Bishop–Phelps property and the G -Bishop–Phelps property. Notice that it is the only instance of the phenomenon, among the properties we have considered here, that the classical property does not obviously imply its G -analogue. It is possibly also related to the apparent difficulty of showing that strong G -dentability implies the G -BP (for a compact group G). It would be certainly very interesting to find an example of G -Banach space with the BP which is strongly G -dentable but fails the G -BP, thereby disproving both implications at once.

2 Question, literature check, and a definition caveat

All Banach spaces below are real. We use the terminology of Dantas–Doucha–Jung–Raunig [1]. Their Problem 8.2 asks, among other dotted arrows, whether

$$G\text{-RNP} \implies \text{RNP} \tag{1.1}$$

holds for every topological group and every continuous isometric action on a Banach space.

The arXiv record of [1] still contained only version 1 when this note was prepared. Targeted searches through June 22, 2026 found no article or preprint resolving (1.1). A later preprint citing [1] uses its Theorem 6.4 but does not address Problem 8.2 [2]. The general fact that exotic groups have no nontrivial nonsingular near actions is known and is also noted explicitly in a recent paper of Avraham-Re'em and Peterzil [3]; the connection with (1.1) below does not appear in the literature located in this search.

There is a minor technical issue in the definitions as printed in [1]. Equation (3.2) asserts that an X -valued vector measure m with merely $|m| \ll \mu$ induces a bounded operator $T_m : L_1(\mu) \rightarrow X$ by $T_m \chi_A = m(A)$. This is false without the stronger bound $|m| \leq C\mu$. For instance, the scalar measure with density $t^{-1/2}$ on $[0, 1]$ is absolutely continuous but does not define a bounded functional on $L_1[0, 1]$. The natural repair is either to impose $|m| \leq C\mu$ in the operator formulation or to change to a dominating control measure. The finite-group proof and the counterexample below are valid under either repair; in the counterexample every quasi-invariant vector measure is actually zero.

We shall use the following direction of the operator formulation, whose proof is unaffected by this caveat.

Lemma 2.1. *If a G -Banach space X has the G -RNP and $S : L_1(\Omega, \mu) \rightarrow X$ is a bounded G -equivariant operator for a continuous positive isometric action of G on $L_1(\mu)$, then S is representable by a function in $L_\infty(\mu; X)$.*

Proof. Set $m(A) = S\chi_A$. For every measurable E and every finite measurable partition (E_j) of E ,

$$\sum_j \|m(E_j)\| \leq \|S\| \sum_j \mu(E_j) = \|S\| \mu(E),$$

so $|m| \leq \|S\|\mu$. Equivariance of S is exactly the required G -quasi-invariance of m . The G -RNP gives a Bochner density F . Since $|m|(A) = \int_A \|F\| d\mu$, the variation bound implies $\|F\| \leq \|S\|$ almost everywhere. Thus $F \in L_\infty(\mu; X)$ and, first on simple functions and then by continuity, $S\varphi = \int \varphi F d\mu$ for every $\varphi \in L_1(\mu)$. $\square$

3 Proof intuition

The finite-group partial works by inducing an arbitrary vector measure to the disjoint union $G \times \Omega$ and then restricting the equivariant density to a positive-measure sheet. For locally compact σ -compact groups, the same idea survives by averaging over a quasi-invariant Haar-equivalent probability measure and then untwisting the density.

The counterexample uses the opposite phenomenon. Some topological groups, including $G = \text{Homeo}_+([0, 1])$, have no nontrivial continuous Hilbert representations. Via the Mazur map, any admissible continuous positive isometric L_1 -action for such a group becomes a Hilbert representation on L_2 , hence must be trivial. Therefore the G -RNP sees only the fixed-point subspace X^G . Taking $X = C_0((0, 1))$ with the natural composition action gives $X^G = \{0\}$, so the equivariant property holds, while X contains c_0 and therefore fails the classical RNP.

4 Verification of the supplied finite-group proof

The supplied argument is correct, including its variation estimate, equivariance computation, and restriction to a positive-measure sheet. For completeness, here is a concise operator version.

Proposition 4.1. *Let G be a finite Hausdorff topological group and let X be a G -Banach space. If X has the G -RNP, then X has the classical RNP.*

Proof. A finite Hausdorff group is discrete. Let $T : L_1(\Omega, \mu) \rightarrow X$ be bounded. Give G normalized counting measure and, for $\Phi \in L_1(G \times \Omega)$, write $\Phi_g(\omega) = \Phi(g, \omega)$. Define

$$\tilde{T}\Phi = \frac{1}{|G|} \sum_{g \in G} g T(\Phi_g).$$

Let G act on $G \times \Omega$ by left translation in the first coordinate. Then $\tilde{T}$ is bounded and G -equivariant. By Lemma 2.1, there is $F \in L_\infty(G \times \Omega; X)$ representing $\tilde{T}$. Fix $g_0 \in G$ and test the representation formula on functions supported on $\{g_0\} \times \Omega$. For every $f \in L_1(\mu)$,

$$\frac{1}{|G|} g_0 T(f) = \frac{1}{|G|} \int_{\Omega} f(\omega) F(g_0, \omega) d\mu(\omega).$$

Consequently

$$T(f) = \int_{\Omega} f(\omega) (g_0^{-1} F(g_0, \omega)) d\mu(\omega).$$

Thus every bounded operator from an L_1 -space into X is representable, which is the classical operator characterization of the RNP. $\square$

Remark 4.2. The vector-measure proof in the supplied packet is equally valid. Its key estimate is

$$|M|(E) \leq \frac{1}{|G|} \sum_{g \in G} |m|(E_g),$$

which follows by applying the triangle inequality to an arbitrary finite partition of E . Since each singleton sheet has positive counting measure, restricting the density to one sheet is legitimate.

5 Two positive extensions

5.1 Weak G -dentability is ordinary dentability

Lemma 5.1. *With Definition 3.5 of [1] exactly as printed, a G -Banach space is weakly G -dentable if and only if it is dentable in the classical sense.*

Proof. Classical dentability plainly implies weak G -dentability. Conversely, assume every bounded G -invariant subset of X is dentable. Let $A \subseteq X$ be a nonempty bounded set and fix $\varepsilon > 0$. Its saturation $G \cdot A$ is bounded and G -invariant, hence dentable. Choose $y \in G \cdot A$ such that

$$y \notin \overline{\text{co}}((G \cdot A) \setminus B_\varepsilon(y)).$$

Write $y = gx$ with $x \in A$. Since g is an isometry,

$$g(A \setminus B_\varepsilon(x)) \subseteq (G \cdot A) \setminus B_\varepsilon(y).$$

If x belonged to $\overline{\text{co}}(A \setminus B_\varepsilon(x))$, applying g would contradict the choice of y . Hence A is dentable. Since A and ε were arbitrary, X is classically dentable. $\square$

Read together with the implication (1) $\Rightarrow$ (3) in Theorem 6.4 and Example 6.2 of [1], this already gives the following consequence.

Corollary 5.2. *Assuming Theorem 6.4 of [1] under the natural definition repair discussed above, if G has property F , then G -RNP implies RNP. In particular, this holds for every locally compact σ -compact group, without a separability assumption.*

5.2 A direct induction theorem

The finite-sheet construction has a continuous analogue. The fact that individual Haar-measure sheets are null is not an obstruction: after obtaining a density on $G \times \Omega$, one untwists and averages that density over G .

Theorem 5.3. *Let G be a locally compact σ -compact group and let X be a G -Banach space. If X has the G -RNP, then X has the classical RNP.*

Proof. Let λ be a left Haar measure on G . Since G is σ -compact, λ is σ -finite. Choose a Borel function $w > 0$ almost everywhere with $\int_G w d\lambda = 1$, and put $d\nu = w d\lambda$. Then ν is a probability measure equivalent to λ . Left translations act nonsingularly on (G, ν) and continuously on $L_1(G, \nu)$ by positive isometries. Indeed, the isometry

$$J : L_1(G, \nu) \longrightarrow L_1(G, \lambda), \quad Jf = wf,$$

conjugates this action to the usual strongly continuous left regular action on $L_1(G, \lambda)$.

Let $T : L_1(\Omega, \mu) \rightarrow X$ be bounded. On $G \times \Omega$ use the product measure $\nu \times \mu$ and the action $h \cdot (g, \omega) = (hg, \omega)$. For $\Phi \in L_1(G \times \Omega)$ define

$$\tilde{T}\Phi = \int_G g T(\Phi_g) d\nu(g). \quad (3.1)$$

The Bochner integral is well defined. One way to see the measurability is that G is a countable union of compact sets, so every norm-continuous orbit in X is separable; approximation of Φ by product-simple functions then gives strong measurability. Moreover,

$$\|\tilde{T}\Phi\| \leq \|T\| \int_G \|\Phi_g\|_{L_1(\mu)} d\nu(g),$$

so $\tilde{T}$ is bounded.

For $h \in G$, let

$$r_h = \frac{d(h_*\nu)}{d\nu}.$$

The induced positive isometry on the product L_1 -space is

$$(U_h\Phi)(g, \omega) = r_h(g)\Phi(h^{-1}g, \omega).$$

Since $r_h d\nu = d(h_*\nu)$, a change of variables gives

$$\begin{aligned} \tilde{T}(U_h\Phi) &= \int_G g r_h(g) T(\Phi_{h^{-1}g}) d\nu(g) \\ &= \int_G hk T(\Phi_k) d\nu(k) = h \tilde{T}\Phi. \end{aligned}$$

Thus $\tilde{T}$ is G -equivariant. Lemma 2.1 supplies $F \in L_\infty(G \times \Omega; X)$ such that

$$\tilde{T}\Phi = \int_{G \times \Omega} \Phi(g, \omega) F(g, \omega) d\nu(g) d\mu(\omega).$$

For $A \in \Sigma$, set

$$q_A(g) = \int_A F(g, \omega) d\mu(\omega).$$

Testing on $\chi_B(g)\chi_A(\omega)$, where $B \subseteq G$ is arbitrary and measurable, yields

$$\int_B q_A(g) d\nu(g) = \int_B g T(\chi_A) d\nu(g).$$

Uniqueness of Bochner-integrable functions therefore gives

$$q_A(g) = g T(\chi_A) \quad \text{for } \nu\text{-almost every } g.$$

Define

$$f(\omega) = \int_G g^{-1} F(g, \omega) d\nu(g).$$

Bochner–Fubini applies and $f \in L_\infty(\Omega; X)$. Hence, for every $A \in \Sigma$,

$$\begin{aligned} \int_A f d\mu &= \int_G g^{-1} q_A(g) d\nu(g) \\ &= \int_G T(\chi_A) d\nu(g) = T(\chi_A). \end{aligned}$$

It follows by linearity and continuity that $T\varphi = \int_\Omega \varphi f d\mu$ for every $\varphi \in L_1(\mu)$. Thus T is representable. Since T and the finite measure space were arbitrary, X has the RNP. $\square$

This theorem contains the supplied finite-group result and also covers every compact group and every countable discrete group.

6 Exotic groups make the equivariant property too weak

Call a topological group *unitarily trivial* if every strongly continuous orthogonal representation on a real Hilbert space is trivial. Such groups are also called exotic. The next observation is the key.

Lemma 6.1 (Mazur-map transfer). *Let G be unitarily trivial. Every continuous action of G on a finite measure space by regular set isomorphisms is trivial modulo null sets. Equivalently, every strongly continuous homomorphism*

$$U : G \longrightarrow \text{Iso}^+(L_1(\mu))$$

is trivial.

Proof. For $f \in L_1(\mu)$ define the Mazur map

$$M(f) = \text{sgn}(f)|f|^{1/2} \in L_2(\mu).$$

By the positive Banach–Lamperti representation, for every $g \in G$ there are a regular set isomorphism θ_g and a nonnegative Radon–Nikodym weight a_g such that

$$U_g f = a_g \tilde{\theta}_g f.$$

Consequently

$$V_g \xi = a_g^{1/2} \tilde{\theta}_g \xi \quad (\xi \in L_2(\mu))$$

is a linear orthogonal operator and

$$M(U_g f) = V_g M(f). \quad (4.1)$$

Thus $(V_g)_{g \in G}$ is an orthogonal representation.

It is strongly continuous. For real scalars s, t ,

$$|\operatorname{sgn}(s)\sqrt{|s|} - \operatorname{sgn}(t)\sqrt{|t|}|^2 \leq 2|s - t|,$$

so

$$\|M(f) - M(k)\|_2^2 \leq 2\|f - k\|_1.$$

For $\xi \in L_2$, put $f = M^{-1}(\xi) = \operatorname{sgn}(\xi)|\xi|^2$. Strong continuity of U and (4.1) imply strong continuity of V . Since G is unitarily trivial, $V_g = I$ for every g . Injectivity of M then gives $U_g = I$. Applying this identity first to 1 and then to characteristic functions shows that $a_g = 1$ and $\theta_g A = A$ modulo null sets for every measurable A . Thus the measure-algebra action is trivial. $\square$

Proposition 6.2. *Let G be unitarily trivial and let X be a G -Banach space. Then*

$$X \text{ has the } G\text{-RNP} \iff X^G := \{x \in X : gx = x \text{ for all } g \in G\} \text{ has the RNP.}$$

Proof. Assume first that X^G has the RNP. By Lemma 6.1, every measure-space action occurring in the definition of the G -RNP is trivial. A G -quasi-invariant vector measure m therefore satisfies $gm(A) = m(A)$ for every g and A , so its range is contained in X^G . The RNP of X^G supplies the required density.

Conversely, assume X has the G -RNP. Equip an arbitrary finite measure space with the trivial G -action. Every X^G -valued absolutely continuous vector measure is then G -quasi-invariant, hence has an X -valued Bochner density F . Applying the quotient map $X \rightarrow X/X^G$ to the integral identity gives

$$\int_A (F + X^G) d\mu = 0 \quad (A \in \Sigma).$$

Uniqueness of Bochner densities implies $F(\omega) \in X^G$ almost everywhere. Hence X^G has the RNP. $\square$

7 The counterexample

Let

$$G = \operatorname{Homeo}_+([0, 1]),$$

the Polish group of orientation-preserving self-homeomorphisms of the closed interval, with the compact-open topology. Let

$$X = C_0((0, 1)) = \{f \in C[0, 1] : f(0) = f(1) = 0\}$$

with the supremum norm, and define

$$(g \cdot f)(t) = f(g^{-1}(t)). \quad (5.1)$$

Theorem 7.1. *The action in (5.1) is a continuous faithful action by linear isometries, X has the G -RNP, and X does not have the classical RNP. Consequently, the implication $G\text{-RNP} \Rightarrow \text{RNP}$ is false for general topological groups.*

Proof. Each operator in (5.1) is a linear isometry. If $g_i \rightarrow g$ in the compact-open topology, then $g_i^{-1} \rightarrow g^{-1}$ uniformly. Since every $f \in X$ is uniformly continuous on $[0, 1]$,

$$\|g_i \cdot f - g \cdot f\|_\infty \rightarrow 0.$$

Thus the representation is strongly continuous, and hence the action is jointly continuous. It is faithful because continuous functions separate points.

Megrelishvili proved that every weakly continuous representation of $\text{Homeo}_+([0, 1])$ by linear isometries on a reflexive Banach space is trivial [4]. In particular, G is unitarily trivial.

Moreover, $X^G = \{0\}$. The group G acts transitively on $(0, 1)$, so an invariant function is constant there; continuity and the endpoint conditions force that constant to be zero. Proposition 6.2 now gives the G -RNP of X .

It remains to show that X does not have the RNP. Choose pairwise disjoint closed intervals $I_n \subset (0, 1)$ accumulating only at 0, and choose tent functions $\varphi_n \in X$ with $\text{supp } \varphi_n \subseteq I_n$ and $\|\varphi_n\|_\infty = 1$. Then

$$J : c_0 \rightarrow X, \quad J((a_n)) = \sum_{n=1}^{\infty} a_n \varphi_n,$$

is a linear isometry. Indeed, the supports are disjoint, the supremum norm is $\sup_n |a_n|$, and continuity at the accumulation point follows from $a_n \rightarrow 0$. Its range is closed. Since the RNP passes to closed subspaces and c_0 fails the RNP, X fails the RNP. $\square$

For completeness, the last standard fact has a short direct proof.

Lemma 7.2. *The Banach space c_0 does not have the RNP.*

Proof. Let (r_n) be the Rademacher functions on $[0, 1]$ and define

$$m(A) = \left(\int_A r_n d\lambda \right)_{n \geq 1}.$$

For every measurable A , Bessel's inequality applied to $\chi_A \in L_2[0, 1]$ shows that the coordinates tend to zero, so $m(A) \in c_0$. Also $\|m(A)\|_\infty \leq \lambda(A)$. This estimate shows countable additivity in the c_0 norm and, for every finite measurable partition (A_j) of A ,

$$\sum_j \|m(A_j)\|_\infty \leq \sum_j \lambda(A_j) = \lambda(A),$$

so $|m| \leq \lambda$.

If m had a Bochner density $F : [0, 1] \rightarrow c_0$, applying the n th coordinate functional would give $F_n = r_n$ almost everywhere. Outside one common null set this holds for every n , so $F(t) = (r_n(t))_{n \geq 1}$. Every coordinate has absolute value 1, and this sequence is not in c_0 , a contradiction. $\square$

8 Conclusion

The supplied finite-group result is valid. The positive statement extends to all locally compact σ -compact groups, both by the direct induction argument above and, after observing Lemma 5.1, through the source paper's property-F theorem. The unrestricted implication fails: for exotic groups the admissible measure-algebra actions can all be trivial, so the G -RNP sees only the fixed-point subspace X^G and can miss the geometry of the rest of X completely.

9 Scope and human review recommendation

The packet is a candidate counterexample to the unrestricted dotted implication $G\text{-RNP} \Rightarrow \text{RNP}$ in Problem 8.2. It is not a negative answer for locally compact σ -compact groups; the packet proves the positive implication in that class. Human review should focus on the definition repair in Section 2, the Mazur-map transfer from positive L_1 -actions to Hilbert representations, the exact form of Megrelishvili’s triviality theorem, and whether the source paper’s intended scope excludes non-locally-compact topological groups.

References

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